

CSA 0378-DATA STRUCTURES

11. STACK USING LINKED LIST

```
main.c
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node
5 {
6     int data;
7     struct node *next;
8 };
9
10 struct node *top = NULL;
11
12 void push(int value)
13 {
14     struct node *newNode;
15
16     newNode = (struct node *)malloc(sizeof(struct node));
17     newNode->data = value;
18     newNode->next = top;
19     top = newNode;
20
21     printf("%d pushed into stack\n", value);
22 }
23
24 void pop()
25 {
26     struct node *temp;
27
28     if(top == NULL)
29     {
30         printf("Stack Underflow\n");
31         return;
32     }
33
34     temp = top;
35     printf("Popped Element: %d\n", temp->data);
36 }
```

Output

```
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 3
```

```
main.c
32 }
33
34 temp = top;
35 printf("Popped Element: %d\n", temp->data);
36
37 top = top->next;
38 free(temp);
39 }
40
41 void display()
42 {
43     struct node *temp;
44
45     if(top == NULL)
46     {
47         printf("Stack is Empty\n");
48         return;
49     }
50
51     printf("Stack Elements:\n");
52
53     temp = top;
54
55     while(temp != NULL)
56     {
57         printf("%d ", temp->data);
58         temp = temp->next;
59     }
60
61     printf("\n");
62 }
63
64 int main()
65 {
66     int choice, value;
67 }
```

Output

```
1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 2
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 3
```

```
main.c
64 int main()
65 {
66     int choice, value;
67
68     while(1)
69     {
70         printf("\n1. Push\n");
71         printf("2. Pop\n");
72         printf("3. Display\n");
73         printf("4. Exit\n");
74         printf("Enter Choice: ");
75         scanf("%d", &choice);
76
77         if(choice == 1)
78         {
79             printf("Enter Value: ");
80             scanf("%d", &value);
81             push(value);
82         }
83         else if(choice == 2)
84         {
85             pop();
86         }
87         else if(choice == 3)
88         {
89             display();
90         }
91         else if(choice == 4)
92         {
93             break;
94         }
95         else
96         {
97             printf("Invalid Choice\n");
98         }
99     }
}
```

Output

```
2 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 pushed into stack

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 3

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 2
Popped Element: 2

1. Push
2. Pop
3. Display
4. Exit
Enter Choice: 3
Stack is Empty

1. Push
2. Pop
3. Display
4. Exit
Enter Choice:
```